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Infotact Technical Internship Program: Advanced Data Analytics Project Specifications and Evaluation Criteria

The following document establishes the technical requirements, analytical directives, and sprint-based data development roadmaps for three distinct Data Analytics (DA) projects. Designed for execution within a fast-paced technology environment in Bengaluru, Karnataka, these specifications serve as comprehensive blueprints for data analytics interns entering the Infotact startup ecosystem.

In 2026, industry expectations for Data Analysts have evolved beyond simple Excel spreadsheets. Modern data teams expect proficiency in end-to-end data pipelines, cloud data warehouses (BigQuery/Snowflake), advanced SQL, Python (Pandas), and the ability to translate raw data into actionable business intelligence. The following projects focus on highly sought-after industry skills: Marketing Attribution, Cohort Retention, and Operational Forecasting.

This document outlines the requisite tasks, technological paradigms, and evaluation criteria for successful internship completion. It is imperative to note that the assessment of these projects relies heavily on the transparency and consistency of the development lifecycle. **The evaluation will only be done if the intern is having all 4 weeks of github commits and contributions.**

Project 1: Multi-Touch Marketing Attribution & ROI Dashboard

Executive Problem Statement

E-commerce and SaaS companies spend millions across multiple advertising channels (Google Ads, Meta, TikTok, LinkedIn). However, customer journeys are complex; a user might click a Facebook ad on Monday, read a blog on Wednesday, and finally convert via a Google Search ad on Friday. Traditional "Last-Click" attribution gives all the credit to Google, leading marketing teams to misallocate their budgets.

The objective of this project is to build a Multi-Touch Attribution model. The intern will ingest fragmented marketing dataset logs, clean the data using SQL/Python, and build an attribution engine that fairly distributes conversion credit across all touchpoints, ultimately calculating the true Return on Ad Spend (ROAS) and Customer Acquisition Cost (CAC).

Business Objectives and Key Performance Indicators

The strategic vision is to stop marketing waste. Success is measured by the analyst's ability to clearly visualize which campaigns drive the highest ROI. The final dashboard must allow marketing managers to toggle between First-Touch, Last-Touch, and Linear attribution models to make data-driven budget allocations.

User Personas and Operational Workflows

Persona	Primary Operational Needs	System Interaction and Workflow
Chief Marketing Officer	Macroscopic visibility on cross-channel ad spend efficiency.	Reviews the dashboard to see true ROAS and reallocates budget from underperforming channels.
Performance Marketer	Granular campaign-level data.	Drills down into specific ad campaigns to see which creatives are driving early-stage funnel awareness vs. bottom-funnel conversions.

Minimum Viable Product Specifications

The foundational feature is the Data Transformation Pipeline. The intern must merge disparate datasets (Ad Spend data + Web Analytics logs + CRM Conversion data) using complex SQL `JOIN`s and Window Functions.
The core deliverable is the interactive Power BI or Tableau dashboard displaying the attribution models, funnel drop-off rates, and dynamic CAC calculations.

Four-Week Engineering Roadmap

Week 1: Data Ingestion and Exploratory Data Analysis (EDA)
The intern will acquire synthetic marketing datasets. Using Python (Pandas) in Jupyter Notebooks, the intern must perform rigorous EDA. Tasks include handling missing UTM parameters, standardizing timestamp formats across different timezone logs, and deduplicating user IDs.

Week 2: Advanced SQL and Attribution Logic
Week two transitions to database logic. The intern will load the cleaned data into a relational database (e.g., PostgreSQL or SQLite) and write advanced SQL queries. The intern must utilize Window Functions (e.g., `ROW_NUMBER() OVER(PARTITION BY user_id ORDER BY timestamp)`) to sequence the user journey and calculate Linear and First-Click attribution weights.

Week 3: Metric Calculation and Data Modeling
The third week focuses on business logic. The intern will calculate the core KPIs: Total Spend, Cost Per Click (CPC), CAC, and ROAS. The data must be modeled into a Star Schema (Fact and Dimension tables) optimized for Business Intelligence tools.

Week 4: BI Dashboarding and Executive Reporting
The final week is dedicated to visualization. The intern will connect Tableau or Power BI to the database. They must build interactive visuals: a conversion funnel, an ROI scatter plot, and a

channel-comparison bar chart. The intern must write an executive summary report translating the data into actionable marketing advice.

Project 2: SaaS/E-Commerce Cohort Retention & CLTV Analysis

Executive Problem Statement

Acquiring new customers is up to five times more expensive than retaining existing ones. High-growth startups often boast about rapid user acquisition but fail to track if those users actually stick around. If a business loses customers faster than it acquires them, it will inevitably collapse.

The objective of this project is to perform a deep-dive Cohort Analysis to understand user retention and calculate Customer Lifetime Value (CLTV). By grouping users based on their acquisition month, the intern will identify exactly when users churn and provide data-backed recommendations to improve retention.

Business Objectives and Key Performance Indicators

The strategic goal is to identify churn patterns. Does a specific product feature cause users to stay? Do users acquired during Black Friday churn faster than others? Success is quantified by producing a clean Cohort Retention Heatmap and an accurate predictive model for the 12-month CLTV of different user segments.

User Personas and Operational Workflows

Persona	Primary Operational Needs	System Interaction and Workflow
Product Manager	Understanding user stickiness and drop-off points.	Analyzes the cohort heatmaps to pinpoint exactly which month after sign-up sees the highest churn, triggering product interventions.
Finance Director	Revenue forecasting and profitability analysis.	Uses the CLTV metrics to determine the maximum acceptable Customer Acquisition Cost (CAC) to remain profitable.

Minimum Viable Product Specifications

The foundational feature is the Cohort Matrix Generation. The intern must use Pandas or SQL to calculate the time difference (in months) between a user's first purchase/login and subsequent activities.

The core deliverable is the analytical notebook and visualization detailing retention curves and segmenting high-value customers from low-value customers.

Four-Week Engineering Roadmap

Week 1: Transactional Data Cleaning and Wrangling

The intern will ingest a raw transactional dataset (e.g., an online retail or SaaS subscription log). Using Python, the intern must filter out refunded/failed transactions, handle missing user IDs, and calculate the "Cohort Month" (the month of the first transaction) for every unique user.

Week 2: Building the Cohort Retention Matrix

Week two is heavily focused on data manipulation. The intern will use Pandas `groupby` and `pivot_table` functions (or SQL pivot techniques) to build a retention matrix. The matrix must calculate the absolute number of retained users and the percentage retention rate for Month 0, Month 1, Month 2, etc.

Week 3: Customer Lifetime Value (CLTV) Calculation

The third week introduces financial metrics. The intern will segment the cohorts (e.g., by acquisition channel or geographic region) and calculate the Average Order Value (AOV) and Purchase Frequency. Using these metrics, the intern will calculate the historical CLTV for each segment.

Week 4: Visualization and Strategic Insights

The final week focuses on storytelling. The intern will use Seaborn/Matplotlib or a BI tool to plot the Retention Heatmap and line charts showing retention decay curves. The final GitHub submission must include a detailed README explaining the business implications (e.g., "Retention drops by 40% in Month 2; recommend implementing an automated re-engagement email sequence").

Project 3: Supply Chain Analytics - Demand Forecasting & Anomaly Detection

Executive Problem Statement

Supply chain efficiency dictates profitability for retail and manufacturing companies. Overestimating demand leads to bloated warehouses, tied-up capital, and spoilage. Underestimating demand leads to stockouts, lost sales, and furious customers. Furthermore, unexpected anomalies (like sudden viral trends or supplier delays) often go unnoticed in massive datasets until the financial damage is already done.

This project requires the intern to build a time-series analytics pipeline. The intern will analyze historical inventory and sales data to forecast future demand, while simultaneously deploying statistical methods to automatically detect anomalies in the supply chain flow.

Business Objectives and Key Performance Indicators

The primary objective is to optimize inventory levels. Success is measured by the accuracy of the forecasting model, quantified by minimizing the Mean Absolute Percentage Error (MAPE) or Root Mean Square Error (RMSE). The anomaly detection system must accurately flag outliers without causing "alert fatigue" from too many false positives.

User Personas and Operational Workflows

Persona	Primary Operational Needs	System Interaction and Workflow
Supply Chain Manager	Data-driven procurement scheduling.	Uses the demand forecast to order the exact amount of raw materials needed for the upcoming quarter.
Operations Analyst	Immediate notification of operational hiccups.	Investigates automated alerts when the anomaly detection system flags an unusual drop in warehouse dispatch rates.

Minimum Viable Product Specifications

The application foundation is the Time-Series Preprocessing module. The intern must resample daily data into weekly/monthly aggregates, handle missing dates using interpolation, and decompose the time series to understand trend and seasonality.
The core deliverable is an interactive web application (built with Streamlit) that allows stakeholders to select a product category, view the next 90 days of forecasted demand, and visually identify historical anomalies on the chart.

Four-Week Engineering Roadmap

Week 1: Time-Series Preprocessing and Decomposition
The intern must acquire a supply chain dataset (e.g., daily sales/inventory logs). Using Pandas, they will set the datetime index and handle missing values. The intern must use statsmodels to decompose the time series, visually separating the underlying trend, seasonal fluctuations, and residual noise.

Week 2: Statistical Anomaly Detection
Week two focuses on identifying outliers. The intern will implement statistical anomaly detection techniques (e.g., Z-Score analysis, Interquartile Range (IQR), or Isolation Forests) to flag days where sales or inventory levels deviated significantly from the expected norm. These anomalies must be contextualized (e.g., Was it a holiday? A stockout?).

Week 3: Demand Forecasting Models
The third week introduces predictive analytics. The intern will establish a baseline forecast using simple moving averages. They will then implement a more robust forecasting algorithm, such as ARIMA or Facebook Prophet. The intern must split the data chronologically (train/test split) to validate the model's accuracy against unseen data.

Week 4: Streamlit Deployment and Final Polish
The final sprint involves transitioning from a static notebook to an interactive application. The intern will use the Streamlit Python library to build a lightweight data app. The app must feature

dropdowns to select different products and sliders to adjust forecast confidence intervals. The app's source code and deployment instructions must be meticulously documented on GitHub.

Universal Data Analytics and GitHub Version Control Guidelines

To simulate a rigorous, enterprise-grade data environment indicative of top-tier product companies in Bengaluru, strict adherence to modern analytical practices and version control is an absolute requirement. The core tenet of this internship program is continuous, transparent, and high-quality contribution. Analysis is not merely the final dashboard; it is the iterative, documented process of problem-solving.

Mandatory Evaluation Protocol

The evaluation will only be done if the intern is having all 4 weeks of github commits and contributions.

A project submitted with a compressed commit history—for example, uploading the entire Jupyter Notebook, all datasets, and the final dashboard in a massive, monolithic push during the final week—will result in immediate disqualification.

Instructions for Tracking Week-Wise Production with Timestamps

Data Analytics requires a specific approach to version control. To pass the evaluation, interns must utilize GitHub's native tools to explicitly track their weekly production.

1. Setup GitHub Projects (Kanban Board)

At the start of Week 1, the intern must navigate to their repository's **Projects** tab and create a new Kanban board (To Do, In Progress, Done).

- **Mandatory Action:** The intern must break down the 4-week roadmap into individual GitHub Issues (e.g., Issue #1: "Clean missing UTM parameters", Issue #2: "Write SQL for Cohort Matrix").
- **Timestamping:** GitHub automatically timestamps when an issue is created, moved, and closed. Managers will audit this board to verify a steady, weekly cadence of data analysis tasks.

2. Commit Frequency and Semantic Issue Referencing

Commits act as the immutable, timestamped ledger of an analyst's thought process. Interns are expected to commit their SQL scripts, Python code, and notebook updates frequently (3-5 times per active development day).

- **Mandatory Action:** Every commit message must reference the specific GitHub Issue it solves using semantic messaging.
- **Format Example:** `eda: identify and remove outlier transactions (fixes #3)` or `sql: implement window function for attribution (fixes #5)`
- By appending `(fixes ##)`, GitHub links the timestamped code commit to the project management board, providing undeniable proof of week-over-week contribution.

3. Managing Jupyter Notebooks in Git

Jupyter Notebooks (`.ipynb`) are notoriously difficult to track in Git because output cells and metadata change constantly.

- **Mandatory Action:** Before committing a Jupyter Notebook, the intern **must clear all cell outputs** (Kernel -> Restart & Clear Output) or use a pre-commit hook like nbstripout. This ensures the Git history tracks the actual *code* changes, not massive blocks of serialized data or chart images.

4. Absolute Data Privacy and `.gitignore` Practices

Security hygiene is paramount. **Pushing raw `.csv`, `.xlsx`, or database `.sqlite` files to GitHub is strictly forbidden and grounds for immediate failure.**

- **Mandatory Action:** The intern must utilize a `.gitignore` file from Day 1 to exclude all `data/` directories, API keys, and database credentials.
- Instead of uploading data, the repository's `README.md` must include a secure link to the dataset or instructions on how to simulate/generate the mock data required to run the analysis scripts.

By strictly enforcing these GitHub tracking specifications, the program ensures that deliverables transcend mere academic exercises, transforming into robust, highly professional Data Analytics portfolios indicative of elite industry standards.